UniPrototype: Human-Robot Skill Learning with Uniform Prototypes

Xiao Hu[1], Yip Fun Yeung[2], Yuhai Wang[1], Tianyu Zhou[3], Fang Xu[3], Jing Du[3], and Gilbert Ye[1,*]

***Abstract*— Data scarcity remains a fundamental bottleneck for robotic manipulation: robots cannot readily leverage the vast amount of human skill data available on the Internet. Bridging this embodiment gap requires transferable skill representations that can be reliably matched across human and robot domains. We propose UniPrototype, a framework that transfers skills by discovering and aligning latent skill representations through prototype assignment across diverse tasks. UniPrototype learns a compositional representation in which multiple prototypes can co-activate, capturing continuous mixtures of reusable skill factors and improving transferability. To further scale with task diversity, we introduce an adaptive prototype selection strategy that automatically adjusts the number of active prototypes based on task complexity, yielding an efficient representation without sacrificing expressiveness. We validate UniPrototype through extensive experiments in simulation and on real-world robotic systems, where it consistently improves task performance and enables effective human-to-robot skill transfer compared to prior approaches.**

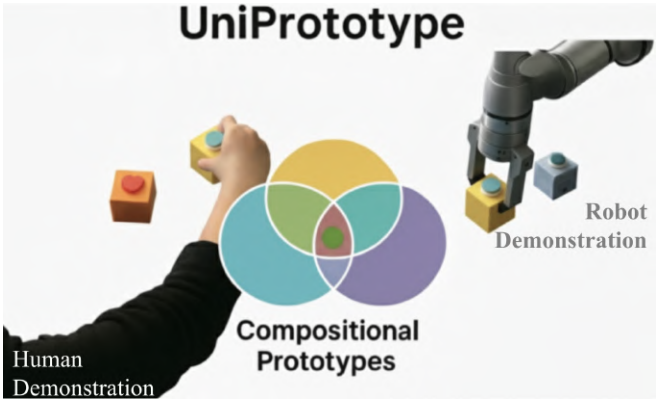


Fig. 1. **UniPrototype** learns a compositional and adaptive prototype representation that bridges the embodiment gap. Multiple prototypes can co-activate to capture transferable skill features, while the prototype set size is automatically adjusted to match task complexity.

I. INTRODUCTION

Data scarcity remains a fundamental bottleneck of robotic manipulation. While humans generate abundant demonstrations in the wild—from instructional videos to everyday interactions—robots cannot readily exploit this vast reservoir for skill learning [1]. The core obstacle is the embodiment gap: humans and robots differ substantially in morphology, kinematics, and action spaces, so the same high-level intent may correspond to very different motions and control signals [2], [3], [6]. As a result, directly transferring skills from human videos to robot policies is nontrivial, and robot learning still relies heavily on expensive on-robot trials.

Significant progress has been made in acquiring features or representations that can (partially) bridge the embodiment gap. A major line of work learns skill representations from visual observations (or learned video features) and then trains a policy to map them to robot actions [1]. For example, XSkill [8] discovers cross-embodiment skill prototypes directly from unlabeled human and robot manipulation video clips and uses the inferred prototype sequence for downstream imitation learning; in contrast, movement-primitive libraries such as DMPs and ProMPs [4], [5] rely on pre-defined, parameterized motion bases and learn to compose primitives to generate robot trajectories. Such end-to-end visual representations are appealing because they are simple, require minimal manual structure, and often work well when large-scale robot data is available. While these studies presented promising results, it is worth diving deeper into the question: Is there a better way to represent skills for cross-embodiment transfer? Visual embedding can be entangled with embodiment-specific appearance and motion details, making it difficult to isolate what is truly transferable across humans and robots. Moreover, these representations are typically explicitly mapped to video clips or images, which contradicts human behaviors where motions are a complex composition of multiple fundamental skills [9].

In fact, many manipulation behaviors can be viewed as compositions of reusable fundamental skills that recur across tasks and embodiments. For example, pick-and-place naturally decomposes into reaching, grasping, lifting, transporting, aligning, and releasing and repositioning—subskills that recur across both human and robot executions. [4]. Real-world skills are rarely expressed as a single fundamental skill at a time. Instead, they are often blended and overlapping—for instance, pouring simultaneously requires holding stability, lifting, and rotation. This observation suggests that a useful skill representation should be compositional: multiple representations should be able to co-activate to represent a single moment of behavior. Built on these insights, transferable skill representations across embodiments should not encode exact motion trajectories or direct visual segments, but instead capture the underlying functional structure of the task. In this study, we call these skill representations 'prototypes'.

In this work, we introduce UniPrototype, a framework that bridges the embodiment gap by automatically discovering and composing continuous skill representations across diverse tasks and embodiments. Our first key idea is a skill-representation discovery mechanism that learns a set

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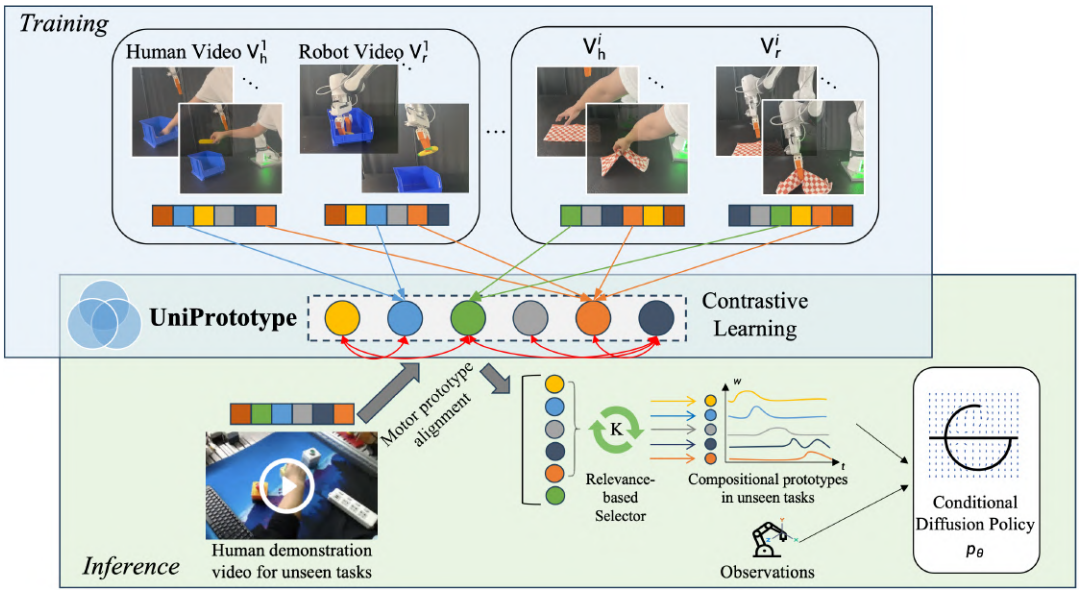


Fig. 2. **Overview** of the UniPrototype framework. Given *unpaired* human and robot demonstrations, UniPrototype learns *compositional* prototype sequences that capture transferable skill primitives across embodiments. Demonstrations are encoded into clip-level skill features, which are assigned to a learnable prototype vocabulary via a soft prototype discovery module. An entropy-based criterion adaptively selects the effective number of active prototypes. A cross-attention module further relates human and robot prototype sequences to encourage consistent cross-embodiment correspondence. At inference time, the learned prototype sequences condition a compositional diffusion policy to generate robot actions aligned with the human plan.

of prototypes and represents each skill instance via multi-prototype activations. Rather than enforcing a hard assignment of each time step to a single prototype, UniPrototype allows multiple prototypes to activate simultaneously, enabling the representation to capture transferable features of skills. This co-activation mechanism better matches the structure of real manipulation and produces prototype factors that generalize across differing embodiments and execution styles.

Our second key idea is that the number of prototypes should not be fixed. Different datasets and tasks exhibit varying levels of compositional complexity: simple reaching behaviors may require only a small vocabulary, whereas long-horizon multi-stage tasks demand a richer set of prototypes. To this end, we propose an adaptive prototype selection strategy that automatically adjusts the number of prototypes based on task complexity, yielding a representation that is both scalable (avoiding unnecessary fragmentation) and efficient (avoiding under-representation). Together, compositional co-activation and adaptive selection produce a robust skill representation suitable for transferring knowledge from abundant human demonstrations to data-scarce robot domains.

We validate UniPrototype through extensive experiments in both simulation and real-world robotic systems, demonstrating consistent improvements in task performance and more effective human-to-robot transfer compared to existing approaches.

The contributions of this paper are threefold:

- We propose a skill representation discovery mechanism with soft, compositional co-activation, allowing multiple prototypes to be active simultaneously to capture transferable features of blended manipulation skills.
- We introduce an adaptive prototype selection strategy that automatically determines the appropriate number of prototypes based on task complexity, yielding a scalable and efficient skill representation.
- We demonstrate in simulation and on real robots that such skill representation (prototype) structure consistently improves task performance, indicating effective transfer of human manipulation knowledge to robots.

II. Related Work

Prototype or Primitive Learning Decomposing complex behaviors into reusable primitives has been a longstanding theme in robot learning. Dynamic Movement Primitives (DMPs) [4] provided an early and influential framework, encoding robot motions as parameterized differential equations adaptable to new contexts. Subsequent extensions, such as ProMPs [5] and Interaction Primitives [14], introduced probabilistic formulations and multi-agent coordination. Beyond these parametric formulations, earlier work also explored extracting behavior vocabularies via dimensionality reduction and clustering in handcrafted feature spaces (e.g., Jenkins & Matarić [18]), offering useful insights but typically without deep representation learning.

With the rise of deep learning, data-driven approaches to primitive discovery from demonstrations (LfD) have become prominent, especially in the low-level sensorimotor space. For example, Yildirim and Ugur [10] proposed Conditional Neural Expert Processes and leveraged entropy to capture multiple, functionally equivalent primitives under varying conditions. Pérez-Dattari and Kober [11] learned *stable* motion primitives by combining imitation with contrastive learning to model consistent dynamics across demonstrations. Related deep sequence modeling frameworks also study temporal abstraction

and compositionality, including VAE-based segmentation for discovering prototype boundaries [15], SPiRL for extracting reusable skills with hierarchical reinforcement learning [16], and CompILE for compositional imitation learning via latent segmentation [17]. Despite impressive progress, many of these methods are primarily evaluated within a single embodiment and often focus on learning *individual* low-level primitives. In contrast, UniPrototype targets *compositional* prototype representations that allow multiple prototypes to co-activate and adaptively scale effective prototype usage with task complexity.

Cross-Embodiment Skill Transfer Transferring skills across different embodiments has long been a central challenge in robotics. Early work by Argall et al. [2] provided a comprehensive survey of correspondence problems in learning from demonstration, highlighting the fundamental difficulties arising from morphological differences between teachers and learners. These challenges have since inspired diverse research directions in cross-embodiment learning.

One line of work seeks to bridge the embodiment gap through representation learning. Osa et al. [3] proposed constructing shared latent spaces for cross-morphology transfer via variational inference, while [13] introduced morphology-agnostic representations using adversarial training to suppress embodiment-specific features. Although effective for action mapping and trajectory retargeting, these approaches largely focus on direct correspondences. In contrast, our method emphasizes compositional prototypes that capture functional equivalences across embodiments.

Complementary efforts leverage temporal correspondence. XIRL [6] employs self-supervised contrastive learning to align human and robot demonstrations, and Temporal Cycle-Consistency Learning (TCC) [7] uses temporal cycle-consistency to establish frame-level mappings. While these strategies facilitate alignment across agents, they rarely address the compositional nature of skills. Building on these advances, [8] introduced a framework that learns skills from human demonstrations. However, it assumes exclusive prototype activation—each step maps to exactly one prototype—limiting its ability to capture smooth transitions and compositional behaviors in manipulation. UniPrototype remove the exclusivity constraint (e.g., Sinkhorn-style balancing) and enabling non-exclusive soft assignments in clustering, allowing multiple prototypes to co-activate and more faithfully represent the compositional structure of manipulation skills.

In contrast to prior work that UniPrototype models skills as compositions of multiple continuous prototypes, enabling smooth transitions and robust transfer across embodiments. Two key innovations drive this capability: (1) compositional prototype discovery, which allows multiple primitives to be active simultaneously, capturing continuous and hierarchical prototype structures; and (2) adaptive prototype selection, which automatically adjusts prototype numbers to dataset complexity, avoiding manual hyperparameter tuning. Together, these advances enable scalable transfer of prototypes from human demonstrations to robot execution without paired data or handcrafted correspondences.

III. METHOD

The key objective of Uniprototype is to learn a *compositional* and *adaptive* skill representation. As human behaviors are naturally composed of multiple fundamental skills, Uniprototype segments latent space prototypes through soft skill discovery that maps demonstrations to prototype clusters to ensure prototype co-activation. Prototypes learned through such a method increase transferability in novel tasks. Furthermore, as tasks with different durations and complexities are naturally comprised of different numbers of prototypes, we further design an adaptive prototype selection strategy that automatically adjusts the number of prototypes based on task complexity, yielding a scalable and efficient representation for downstream policy learning and flexible task execution.

A. Compositional Prototype Learning

We introduce a self-supervised learning framework that simultaneously learns temporal prototype representations and their prototype assignments.

a) Temporal Skill Encoding.: Given demonstration videos from humans $\mathcal{S}^h$ and robots $\mathcal{S}^r$, we extract temporal prototype representations that are invariant to embodiment differences. For each video U_n, we uniformly sample N frames and construct overlapping video clips $\{u_{n\ell}\}_{\ell=0}^{N}$ using a sliding window of length W. Each clip captures local motion patterns that correspond to prototype executions. We encode these clips using a temporal encoder g_φ consisting of a vision backbone followed by a transformer architecture:

$$e_{n\ell} = g_\varphi(u_{n\ell})$$

where $e_{n\ell} \in \mathbb{R}^{d_e}$ represents the prototype embedding for clip $u_{n\ell}$. To enhance cross-embodiment alignment, we apply augmentations $\mathcal{A}$ including random crops, color jittering, and geometric transformations.

b) Compositional Prototype Discovery.: Unlike traditional clustering approaches that enforce exclusive cluster assignments, we recognize that manipulation skills naturally exhibit compositional structure. A pick-and-place action, for instance, simultaneously activates the picking, lifting, and subsequently the placing primitives. Note that we use the word 'picking' here to describe a prototype just for demonstration purposes; real prototypes can contain nuanced motions without clear semantic tags. To capture this compositionality, we introduce learnable prototypes $P = [p_1, \ldots, p_{K_p}] \in \mathbb{R}^{d_e \times K_p}$ and design a soft assignment mechanism that allows multiple prototype activations. The projection of prototype representations onto prototypes is computed as:

$$R = P^\top E$$

where $E = [e_1, \ldots, e_{N_b}]$ contains prototype representations from a batch. We then apply an asymmetric normalization that preserves compositional patterns:

$$U = \text{RowNorm}\left(\exp\left(\frac{R}{\lambda}\right)\right)$$

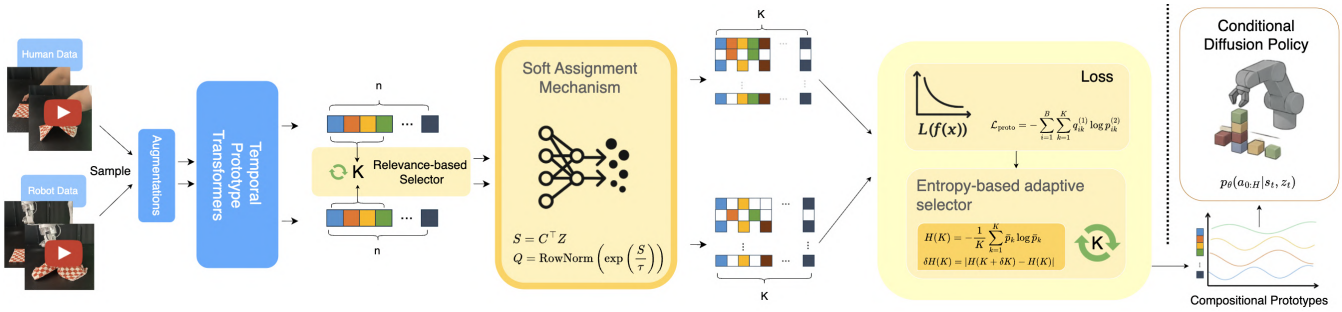


Fig. 3. **Training flow of UniPrototype.** Given human and robot demonstrations, the framework augments raw data and encodes them using temporal prototype transformers. A soft assignment mechanism discovers compositional prototypes, allowing multiple primitives to activate simultaneously. An entropy-based adaptive selector automatically determines the appropriate size of prototypes. The learned prototype assignments are optimized through a self-supervised loss, ensuring temporal coherence and compositional consistency. Finally, the discovered prototypes condition a diffusion-based policy, which generates executable robot actions via iterative denoising, enabling the robot to successfully complete cross-embodiment manipulation tasks.

This yields a soft distribution over prototypes, so multiple prototypes can co-activate within the same clip. The temperature parameter λ controls the sharpness of assignments.

c) Adaptive Prototype Selection.: The number of prototypes K_p critically affects representation quality. Too few prototypes under-represent diversity, while too many lead to fragmentation. We introduce an information-theoretic approach to automatically determine optimal K_p. We define assignment entropy as:

$$\mathcal{H}(K_p) = -\frac{1}{K_p}\sum_{k=1}^{K_p} \bar{u}_k \log \bar{u}_k$$

where $\bar{u}_k$ represents the average activation probability of prototype k across the dataset. During preliminary training with varying K_p, we monitor the rate of entropy change:

$$\Delta\mathcal{H}(K_p) = |\mathcal{H}(K_p + \Delta K) - \mathcal{H}(K_p)|$$

The optimal number of prototypes $K^\star$ is selected when $\Delta\mathcal{H}(K_p) < \epsilon$, indicating that additional prototypes no longer capture meaningful patterns. This data-driven approach adapts to the inherent complexity of the demonstration dataset.

d) Self-Supervised Objective.: We train the temporal encoder and prototypes jointly using a contrastive learning framework [22]. For each video clip, we generate two augmented views and compute their prototype assignments. The learning objective maximizes agreement between different views of the same clip:

$$\mathcal{L}_{\text{assign}} = -\sum_{i=1}^{N_b}\sum_{k=1}^{K_p} u_{ik}^{(a)} \log u_{ik}^{(b)}$$

where $u^{(a)}$ and $u^{(b)}$ are prototype assignments from two augmented views. This encourages consistent prototype activation regardless of visual variations. Additionally, we incorporate temporal coherence through a time-contrastive objective:

$$\mathcal{L}_{\text{tcn}} = -\sum_i \log \frac{\exp(\text{sim}(e_i, e_{i+\Delta})/\gamma)}{\sum_j \exp(\text{sim}(e_i, e_j)/\gamma)}$$

where Δ defines temporal proximity. This ensures that temporally adjacent clips share similar prototype activations.

B. Learning Policies

With discovered compositional prototypes, we train prototype-conditioned policies that map prototype activations to robot actions. We employ a diffusion-based policy [19] architecture that naturally handles the multimodal action distributions arising from compositional skills. The policy $\pi_\omega(u_t|x_t, r_t)$ conditions on both the current state x_t (including proprioception and visual observation) and the compositional prototype representation r_t. Since r_t may activate multiple prototypes, the policy learns to execute blended primitives and smooth transitions. The diffusion process models the conditional action distribution:

$$p_\psi(u_{0:T}|x_t, r_t) = \int p(u_T) \prod_{t'=1}^{T} p_\psi(u_{t'-1}|u_{t'}, x_t, r_t)\, du_{1:T}$$

where T denotes the diffusion horizon. The reverse diffusion process is learned through denoising score matching on robot demonstration data.

The compositional prototype representation offers several key advantages for policy learning. First, smooth transitions emerge naturally from multiple prototype activations, enabling fluid motion blending between primitives. Second, flexible recomposition allows the policy to generate novel behaviors by combining learned prototypes in new ways, enhancing generalization beyond the training distribution. Third, robust alignment through partial prototype matches enables the policy to recover from errors and adapt to variations in task execution. Finally, efficient representation via compositional encoding significantly reduces the dimensionality of the prototype space while maintaining expressiveness.

C. Flexible Task Execution

During inference, *UniPrototype* enables one-shot imitation from human demonstrations of novel tasks. Given a human prompt video ξ_{demo}, we first extract the compositional prototype sequence:

$$\tilde{R} = \{r_t\}_{t=0}^{T_{\text{demo}}} = g_\varphi(\xi_{\text{demo}})$$

Each r_t represents a potentially compositional prototype state with multiple active prototypes. This forms a hierarchical

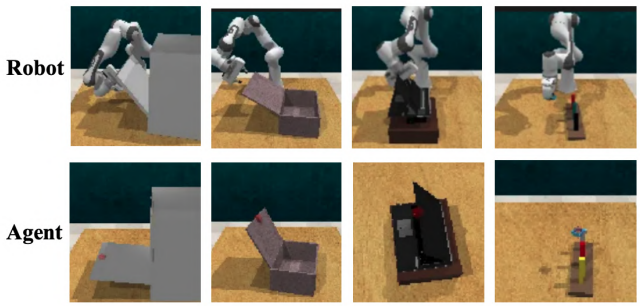


Fig. 4. **Cross-embodiment transfer** in simulation across diverse manipulation tasks [23]: UniPrototype demonstrates robust skill transfer between robot and humanoid agent embodiments. **Top row:** Robot manipulator executing four distinct tasks - empty dishwasher, close box, close laptop lid, and insert into a square peg. **Bottom row:** Humanoid agent performing the same sequence of tasks using transferred compositional prototypes.

task plan where high-level task structure emerges from prototype compositions.

To handle embodiment differences and execution variations, we use cross-attention to match the current robot state with the appropriate position in the compositional prototype sequence:

$$\hat{r}_t = \mathrm{XAttn}\left(y_t^{\text{robot}}, \tilde{R}\right). \tag{1}$$

Here, y_t^{robot} serves as the query, while $\tilde{R}$ provides the matched representation, yielding an aligned prototype representation $\hat{r}_t$ for execution. The attention mechanism identifies which prototypes have been completed and which should be activated next, enabling robust execution despite speed differences and potential failures. The aligned compositional representation $\hat{r}_t$ is then passed to the learned policy for action generation:

$$u_t = \pi_\omega(x_t, \hat{r}_t)$$

Through these innovations, *UniPrototype* achieves effective cross-embodiment prototype transfer while maintaining the flexibility to compose learned primitives to execute novel tasks.

IV. Experiments

We evaluate UniPrototype on a diverse set of manipulation tasks to assess whether such skills representation mitigates the embodiment gap. Following the core design principles of UniPrototype, we structure our experimental analysis to answer the following four questions: (1) **Soft alignment:** does soft, non-exclusive alignment yield a better cross-embodiment skill representation? (2) **Adaptive K:** can the representation automatically scale its prototype capacity with task complexity? (3) **Compositionality:** Can learned skills representation be composed in novel tasks? (4) **Overall effectiveness:** do these components lead to consistent gains over strong baselines in simulation and on real robots?

A. Experimental Setup

a) Environments and evaluation protocol: We use RLBench [21] with the cross-embodiment transfer setting in [23], covering 100 manipulation tasks. We report success rates for both same-embodiment and cross-embodiment transfer, and stress-test transfer under execution-speed mismatches ($\times 1.0$ and $\times 2.0$). For analysis, we heuristically group tasks into four complexity tiers: **Simple** (single-stage reach/push/pick), **Tool-use** (tool or sustained contact, e.g., wipe/sweep/scoop), **Multi-stage** (ordered sub-goals, e.g., open–pick–place–close), and **Complex** (long-horizon and/or precision-heavy, e.g., puzzle/chess/jenga/stacking).

b) Baselines.: We compare UniPrototype against: (1) **GCD Policy**, a goal-conditioned diffusion policy [19]; (2) **GCD Policy w. TCN** [22], which augments GCD with a pre-trained time-contrastive representation; and (3) **XSkill** [8], a representative skill discovery and transfer baseline.

B. Soft Alignment

A central challenge in cross-embodiment transfer is that the same functional intent can unfold with different timing and execution styles. UniPrototype addresses this by using soft alignment as a skill representation: instead of enforcing one-to-one matching or exclusive assignments, each timestep is represented by a distribution over prototypes, allowing partial matches and smooth transitions across embodiments.

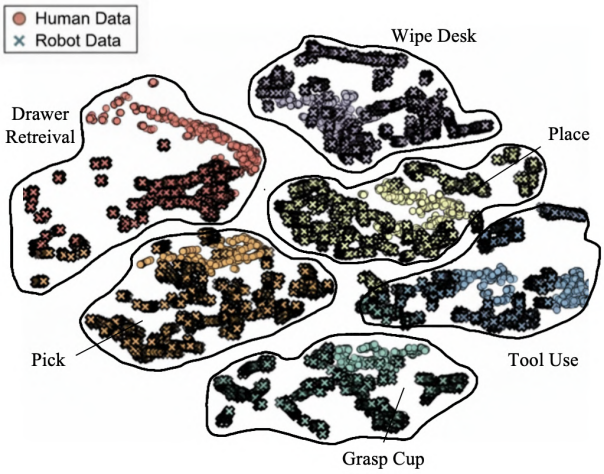


Fig. 5. **t-SNE visualization** of skill features extracted from human demonstrations and robot executions (real-world). Soft alignment yields semantically coherent clusters across skill categories [20], where human (circles) and robot (crosses) overlap within each cluster, indicating effective cross-embodiment representation learning.

a) Consistent cross-embodiment representations.: To visualize this property, Fig. 5 shows a t-SNE projection [25] of the learned skill features from both human demonstrations and robot executions. We observe semantically coherent clusters corresponding to manipulation skill categories, where human (circles) and robot (crosses) instances overlap within the same cluster. This indicates that soft alignment preserves *functional similarity* across embodiments while remaining invariant to embodiment-specific appearance and motion details.

b) Temporal correspondence.: Beyond global feature geometry, UniPrototype also produces temporally meaningful *soft assignments*. Fig. 6 visualizes prototype assignment distributions over time for human demonstrations and robot executions. Multiple prototypes can be partially active around

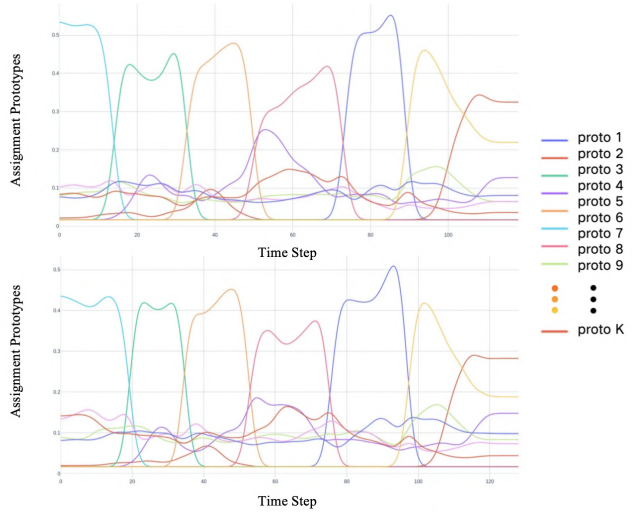


Fig. 6. **Prototype assignment distributions (soft alignment) over time** for human (top) and robot (bottom) trajectories. UniPrototype represents each timestep with a soft distribution over prototypes, producing smooth transitions and consistent cross-embodiment correspondence.

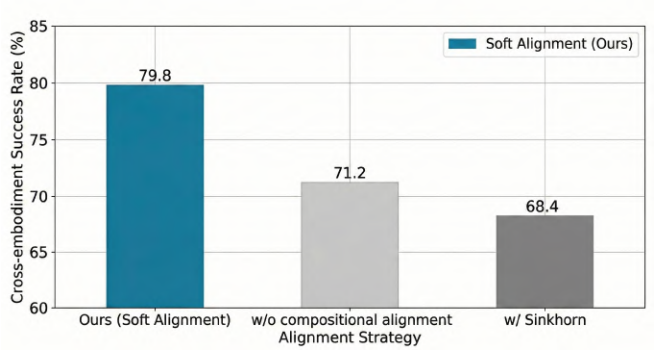


Fig. 7. **Alignment ablation.** Soft alignment outperforms removing compositional alignment and Sinkhorn-based exclusive assignment on cross-embodiment success rate in over 100 manipulation tasks.

transitions, and the overall activation patterns remain aligned across human and robot data. This provides qualitative evidence that soft alignment yields a smooth representation that supports stable cross-embodiment matching.

c) Alignment ablation: Fig. 7 shows that our soft alignment achieves the best cross-embodiment success (**79.8**%). Removing compositional alignment drops performance to **71.2**%, and enforcing exclusive assignments with Sinkhorn further reduces it to **68.4**%. These results indicate that non-exclusive soft alignment is crucial for robust cross-embodiment transfer.

C. Adaptive Prototype Selection

A transferable skill representation should be adaptive: different tasks require different prototype capacities. UniPrototype therefore selects the prototype count K based on task complexity, avoiding both *under-representation* (too few prototypes to express the behavior) and *fragmentation* (too many prototypes that split coherent skills).

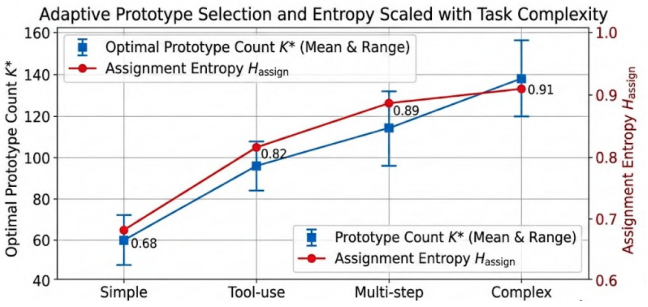


Fig. 8. Optimal prototype count (K^*, left axis, blue squares showing the adaptive range) and assignment entropy (H_{assign}, right axis, red circles) across task categories. As complexity increases from *Simple* to *Complex*, UniPrototype selects larger K^* and exhibits higher entropy, indicating richer and more distributed prototype usage.

a) Match task complexity.: As visualized in Fig. 8, the selected K^* increases with task complexity: *Simple* tasks prefer compact vocabularies ($K \in [48, 72]$), while *Complex* tasks require richer prototype sets ($K \in [120, 156]$). This trend aligns with our motivation that long-horizon behaviors exhibit higher compositional complexity and thus demand more representational capacity.

We use assignment entropy H_{assign} to quantify how broadly the model distributes soft assignments over prototypes: low entropy implies near-exclusive usage of a few prototypes, while high entropy indicates more distributed (mixture-like) activations. In Fig. 8, H_{assign} rises alongside the adaptively selected K^* from *Simple* to *Complex* tasks, suggesting that increased capacity is not merely added but *actively used* to represent richer sub-behaviors and longer-horizon transitions; otherwise, assignments would remain concentrated and entropy would not increase.

b) Fixed K is suboptimal across heterogeneous tasks.: Using a single prototype budget for all tasks cannot simultaneously fit both simple and long-horizon behaviors. Consistently, fixing K (e.g., **w/ fixed** K=128) underperforms adaptive selection. Fig. 9 further illustrates the trade-off: small K (e.g., 32) severely under-represents complex tasks, whereas overly large K (e.g., 256/512) can fragment simpler skills and provide diminishing returns for tasks whose estimated optimal range is $[120, 156]$.

D. Compositionality in Novel Tasks

To directly evaluate whether UniPrototype learns *compositional* and reusable skill factors, we design a novel recombination experiment that requires transferring and *stitching* primitives across tasks and embodiments. During training, we collect human and robot demonstrations for two base tasks: (i) *wipe_desk* and (ii) *pick_banana*. At inference time, we provide a *human* demonstration of a novel task, *wipe_banana*, which is *not* observed during training. The robot is then tasked to execute *wipe_banana* solely from this human video.

UniPrototype succeeds in this setting: the robot performs *wipe_banana* by recombining learned prototypes corresponding to (a) wiping motions and (b) grasping and stabiliza-

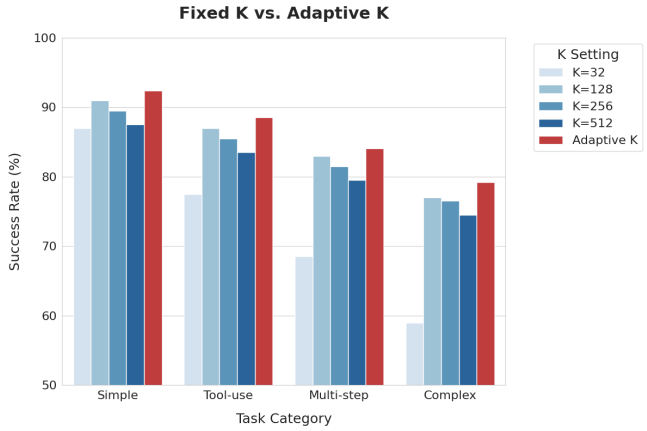


Fig. 9. Illustration of the fixed-K vs. adaptive-K trade-off across task categories: small K under-represents complex tasks, while overly large K fragments simple skills and yields diminishing returns beyond an optimal range. In contrast, adaptive-K selects an appropriate vocabulary size per task, avoiding costly sweeps over multiple fixed-K settings.

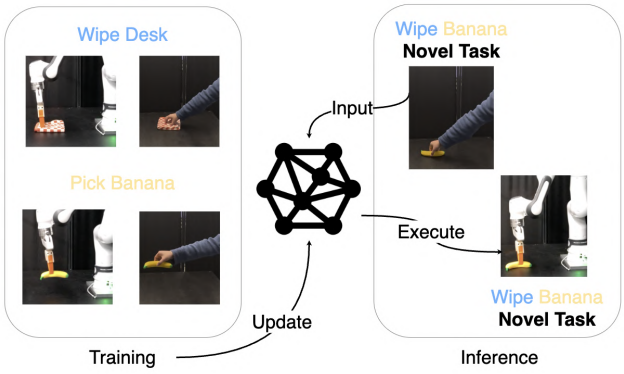


Fig. 10. **Compositional recombination for novel task transfer. Training (left):** UniPrototype is trained with human and robot demonstrations of two base tasks, *wipe_desk* and *pick_banana*. **Inference (right):** given a human demonstration of the unseen composite task *wipe_banana*, UniPrototype recombines the learned wiping and object-centric prototypes to generate a transferable skill representation, enabling the robot to execute *wipe_banana* zero-shot.

tion behaviors. This demonstrates that the learned skill representation is not a flat, task-specific embedding, but a compositional basis in which multiple prototypes can co-activate and be reused across tasks. Crucially, the novel task requires transferring the wiping prototype from *wipe_desk* while grounding it to a new object and interaction geometry borrowed from *pick_banana*. The resulting behavior indicates that UniPrototype captures transferable functional structure and supports zero-shot recombination of skill components across embodiments.

E. Real-World Robot Experiments

To validate practical applicability, we conduct real-world experiments using a Franka Research3 across five manipulation tasks: (1) Table Wiping, (2) Bottle Grasping, (3) Drawer Retrieval, (4) Tool Use, (5) Object placement.

UniPrototype achieves a **77.1%** average success rate, substantially outperforming XSkill (**54.2%**) and GCD Policy (**26.1%**). Gains are particularly strong on tasks requiring continuous transitions (e.g., wiping) and coordinated multi-stage behaviors (e.g., drawer retrieval), consistent with the benefits of soft alignment and adaptive prototype capacity. x

TABLE I

REAL-WORLD SUCCESS RATES (%) OVER 24 TRIALS PER TASK.

Task	GCD Policy	XSkill	UniPrototype
Table Wiping	20.8±4.2	45.8±5.1	**70.8±4.5**
Bottle Grasping	41.7±4.8	66.7±4.7	**83.3±3.7**
Drawer Retrieval	16.7±3.8	50.0±5.1	**75.0±4.4**
Tool Use	25.0±4.4	54.2±5.1	**79.2±4.1**
Average	26.1±4.3	54.2±5.0	**77.1±4.2**

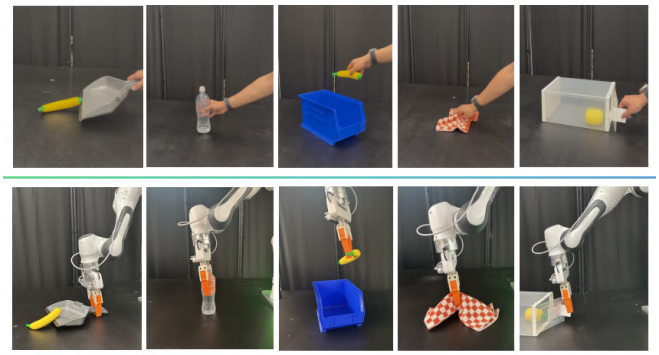


Fig. 11. **Real-world manipulation tasks.** Top row: Human demonstrations of five distinct tasks - spatula flipping, bottle grasping, object placement in bin, table wiping, and drawer retrieval. Bottom row: Robot execution of the same tasks using learned compositional prototypes. UniPrototype successfully transfers these varied manipulation skills.

F. Overall Effectiveness in Simulation and Real Robots

We now evaluate whether soft alignment and adaptive K translate into consistent end-to-end improvements over strong baselines.

TABLE II

CROSS-EMBODIMENT PERFORMANCE

Method	Same	Cross-Embodiment	
		×1.0	×2.0
GCD Policy	68.3 ± 2.1	12.4 ± 1.8	4.1 ± 0.9
GCD Policy w. TCN	71.2 ± 1.9	24.7 ± 2.3	11.6 ± 1.7
XSkill	84.6 ± 1.5	78.2 ± 1.8	52.3 ± 2.4
UniPrototype	91.3 ± 1.2	87.5 ± 1.4	71.2 ± 2.0

a) Simulation transfer performance.: Table II shows that UniPrototype consistently outperforms all baselines under both same-embodiment and cross-embodiment settings, and remains robust under speed mismatch. Importantly, transfer degrades gracefully as the speed gap increases, suggesting that the learned representation captures functional structure beyond embodiment-specific timing.

b) Performance vs. task complexity.: Table III breaks down cross-embodiment transfer by task category. UniPrototype exhibits the largest improvements on *multi-step* and *complex* tasks, where robust transfer requires both flexible alignment and sufficient prototype capacity, consistent with our soft-alignment and adaptive-K analyses.

TABLE III
TASK-SPECIFIC CROSS-EMBODIMENT PERFORMANCE COMPARISON

Task Category	GCD Policy	GCD w. TCN	XSkill	**UniPrototype**
Simple	18.6	31.4	85.7	**92.4**
Tool-use	11.2	25.8	78.3	**88.5**
Multi-step	8.4	19.6	72.4	**84.1**
Complex	5.3	14.2	66.9	**79.2**

V. DISCUSSION AND CONCLUSION

We propose UniPrototype, a framework for discovering and aligning prototypes from unlabeled human and robot demonstrations. UniPrototype shows that compositional prototypes form an effective abstraction for cross-embodiment skill transfer. By enabling simultaneous activations, they capture transitions and continuous motions, yielding strong performance on complex tasks. The entropy-based selection further adapts prototype vocabulary to task complexity, reducing manual effort and improving scalability. When combined with diffusion-based policies, compositional prototypes enhance robustness in both simulation and real-world settings, particularly under speed variations and perturbations. Remaining challenges include additional training costs, higher memory usage, and evaluation beyond structured environments.

While UniPrototype demonstrates strong performance, several limitations remain: (1) The assignment entropy criterion requires preliminary training passes, increasing computational cost. (2) Current experiments use structured lab settings; performance in unstructured environments remains unexplored. (3) The compositional representation increases memory requirements compared to single-prototype methods.

Future work will target unstructured settings, interpretable prototype grounding, and online refinement. Overall, UniPrototype re-frames imitation learning beyond trajectory matching toward scalable human-to-robot skill transfer.

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